

Derivation of Nontrivial LICI Geometries

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1 Introduction

This document presents the mathematical derivation for finding nontrivial Light-Induced Conical Intersection (LICI) geometries in a three-molecule system. The system consists of three identical diatomic molecules placed in a cavity with their axes parallel to the x , y , and z axes of a Cartesian coordinate system.

2 Model Potentials

The system is described by two potential energy surfaces:

$$V_x(x) = x^2 \tag{1}$$

$$V_a(x) = a \cdot (x - x_{\text{shift}})^2 + c_0 \tag{2}$$

Where:

- a is a scaling parameter
- x_{shift} is a shift parameter for the second potential
- c_0 is a constant energy offset

3 Potential Difference Analysis

The difference between the two potentials is:

$$V_a(x) - V_x(x) = a \cdot (x - x_{\text{shift}})^2 + c_0 - x^2 \tag{3}$$

$$= (a - 1) \cdot x^2 - 2 \cdot a \cdot x_{\text{shift}} \cdot x + c_1 \tag{4}$$

Where $c_1 = a \cdot x_{\text{shift}}^2 + c_0$.

This can be rewritten in the form of a perfect square plus a constant:

$$V_a(x) - V_x(x) = (a - 1) \cdot (x - x_{\text{prime}})^2 + c'_1 \tag{5}$$

Where x_{prime} is the point where the quadratic term has its minimum:

$$x_{\text{prime}} = \frac{a}{a - 1} \cdot x_{\text{shift}} \tag{6}$$

4 Trivial and Nontrivial LICIs

4.1 Trivial LICIs

Trivial LICIs occur when all three bond distances are equal:

$$(r_1, r_2, r_3) \text{ with } r_1 = r_2 = r_3 (= r_0) \quad (7)$$

These form a seam with the property $r_1 = r_2 = r_3$.

4.2 Nontrivial LICIs

To find nontrivial LICIs, we search for points where two different bond distances $r_1 \neq r_2$ give equal potential differences:

$$(V_a - V_x)(r_1) = (V_a - V_x)(r_2) \quad (8)$$

If such $r_1 \neq r_2$ exist, the following configurations provide nontrivial LICIs:

$$\text{LICI}_1 : (r_1, r_2, r_3) = (r_1, r_1, r_2) \quad (9)$$

$$\text{LICI}_2 : (r_1, r_2, r_3) = (r_1, r_2, r_1) \quad (10)$$

$$\text{LICI}_3 : (r_1, r_2, r_3) = (r_2, r_1, r_1) \quad (11)$$

$$\text{LICI}_4 : (r_1, r_2, r_3) = (r_2, r_2, r_1) \quad (12)$$

$$\text{LICI}_5 : (r_1, r_2, r_3) = (r_2, r_1, r_2) \quad (13)$$

$$\text{LICI}_6 : (r_1, r_2, r_3) = (r_1, r_2, r_2) \quad (14)$$

5 Geometric Constraints

Since the trivial LICIs form a seam, we set up our closed paths in the plane perpendicular to this "trivial" seam. All points in this plane have the property:

$$r_1 + r_2 + r_3 = 3 \cdot r_0 \quad (15)$$

where (r_0, r_0, r_0) is the trivial LICI in the plane.

It is easy to see that LICI_1 , LICI_2 , and LICI_3 lie in the same plane, and similarly, LICI_4 , LICI_5 , and LICI_6 also lie in the same plane.

6 Derivation for LICI_1 , LICI_2 , LICI_3

We now search for nontrivial LICIs of type LICI_1 , LICI_2 , LICI_3 with the additional requirement:

$$r_1 + r_1 + r_2 = 3 \cdot r_0 \quad (16)$$

for a predefined r_0 .

From the equality condition $(V_a - V_x)(r_1) = (V_a - V_x)(r_2)$ with $r_1 \neq r_2$, we obtain:

$$r_1 = x_{\text{prime}} - \delta, \quad r_2 = x_{\text{prime}} + \delta \quad (17)$$

where δ is an arbitrary parameter.

Applying the geometric constraint:

$$r_1 + r_1 + r_2 = 3 \cdot r_0 \quad (18)$$

$$3 \cdot x_{\text{prime}} - \delta = 3 \cdot r_0 \quad (19)$$

$$\delta = 3 \cdot (x_{\text{prime}} - r_0) \quad (20)$$

Substituting back to find r_1 and r_2 :

$$r_1 = x_{\text{prime}} - \delta = 3 \cdot r_0 - 2 \cdot x_{\text{prime}} = r_0 - 2 \cdot (x_{\text{prime}} - r_0) \quad (21)$$

$$r_2 = x_{\text{prime}} + \delta = 4 \cdot x_{\text{prime}} - 3 \cdot r_0 = r_0 + 4 \cdot (x_{\text{prime}} - r_0) \quad (22)$$

7 Distance from Trivial LICI

To determine how many LICIs are enclosed by a given circle, we need to know the distance between the nontrivial and trivial LICIs. The distance in the 3D coordinate space is:

$$d_{\text{CI}} = |r_2 - r_0| \cdot \frac{\sqrt{6}}{2} \quad (23)$$

$$= 4 \cdot |x_{\text{prime}} - r_0| \cdot \frac{\sqrt{6}}{2} \quad (24)$$

$$= 2 \cdot \sqrt{6} \cdot |x_{\text{prime}} - r_0| \quad (25)$$

8 Analysis of LICI₄, LICI₅, LICI₆

Similar expressions could be derived for LICI₄, LICI₅, and LICI₆. In this case, we would obtain:

$$\delta' = -3 \cdot (x_{\text{prime}} - r_0) = -\delta \quad (26)$$

Taking this value and substituting into the expressions for r_1 and r_2 :

$$r'_1 = x_{\text{prime}} - \delta' = x_{\text{prime}} + \delta = r_2 \quad (27)$$

$$r'_2 = x_{\text{prime}} + \delta' = x_{\text{prime}} - \delta = r_1 \quad (28)$$

Combining this with the definitions of the nontrivial LICI_{*i*} configurations, we realize that these LICIs are actually the same as LICI₁, LICI₂, and LICI₃. More specifically:

$$\text{LICI}_4 = \text{LICI}_1 \quad (29)$$

$$\text{LICI}_5 = \text{LICI}_2 \quad (30)$$

$$\text{LICI}_6 = \text{LICI}_3 \quad (31)$$

9 Conclusion

The derivation shows that for a given plane defined by $r_1 + r_2 + r_3 = 3 \cdot r_0$, there are exactly three distinct nontrivial LICI configurations. These are symmetrically arranged around the trivial LICI at (r_0, r_0, r_0) , with the distance from the trivial LICI given by:

$$d_{\text{CI}} = 2 \cdot \sqrt{6} \cdot |x_{\text{prime}} - r_0| \quad (32)$$

This geometric understanding is crucial for determining the topological properties of the system and for setting up appropriate closed paths for geometric phase calculations.